

TURBINE HEALTH MONITORING AND FAILURE FORECASTING

-A.S.S.V.AKSHAI

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INTRODUCTION

Wind turbines are an important source of renewable energy, but continuous operation can lead to mechanical faults and unexpected failures. These failures may cause downtime, increased maintenance costs, and reduced power generation.

The project “Turbine Health Monitoring and Failure Forecasting” aims to develop a machine learning-based system that monitors the health of wind turbines and predicts possible failures in advance. The model analyses key parameters such as Wind Speed, Rotor Speed, Blade Angle, Gearbox Temperature, Generator Temperature, and Vibration to identify abnormal conditions.

By predicting faults like gearbox, generator, or blade failures before they occur, the system helps in timely maintenance, reduces downtime, and improves the overall efficiency and reliability of wind turbines.

PROBLEM STATEMENT

Wind turbines operate continuously under varying environmental and mechanical conditions, which can lead to faults in critical components such as the gearbox, generator, and blades. Unexpected failures in these components result in increased downtime, reduced energy production, and high maintenance costs.

The project addresses the following key questions:

- Predict specific types of failures (e.g., gearbox, generator, blade) before they occur.
- How can the health of a wind turbine be continuously monitored using operational parameters?
- How can sensor data such as Wind Speed, Rotor Speed, Blade Angle, Gearbox Temperature, Generator Temperature, and Vibration be analysed effectively?

OBJECTIVES

The goal of this project is to develop a predictive model for monitoring the health of wind turbines and forecasting potential failures. By analyzing key operational parameters such as WindSpeed, RotorSpeed, BladeAngle, GearboxTemperature, GeneratorTemperature, and Vibration, we aim to:

- Identify patterns or conditions that lead to failures.
- Predict specific types of failures (e.g., gearbox, generator, blade) before they occur.
- Improve turbine uptime and optimize maintenance schedules, reducing downtime and operational costs.

METHODOLOGY

The methodology outlines the step-by-step process followed in the project, from data collection to final analysis and Prediction. It ensures a structured approach to achieve the project's objectives by using various tools and techniques.

1. Data Exploring and Preprocessing:

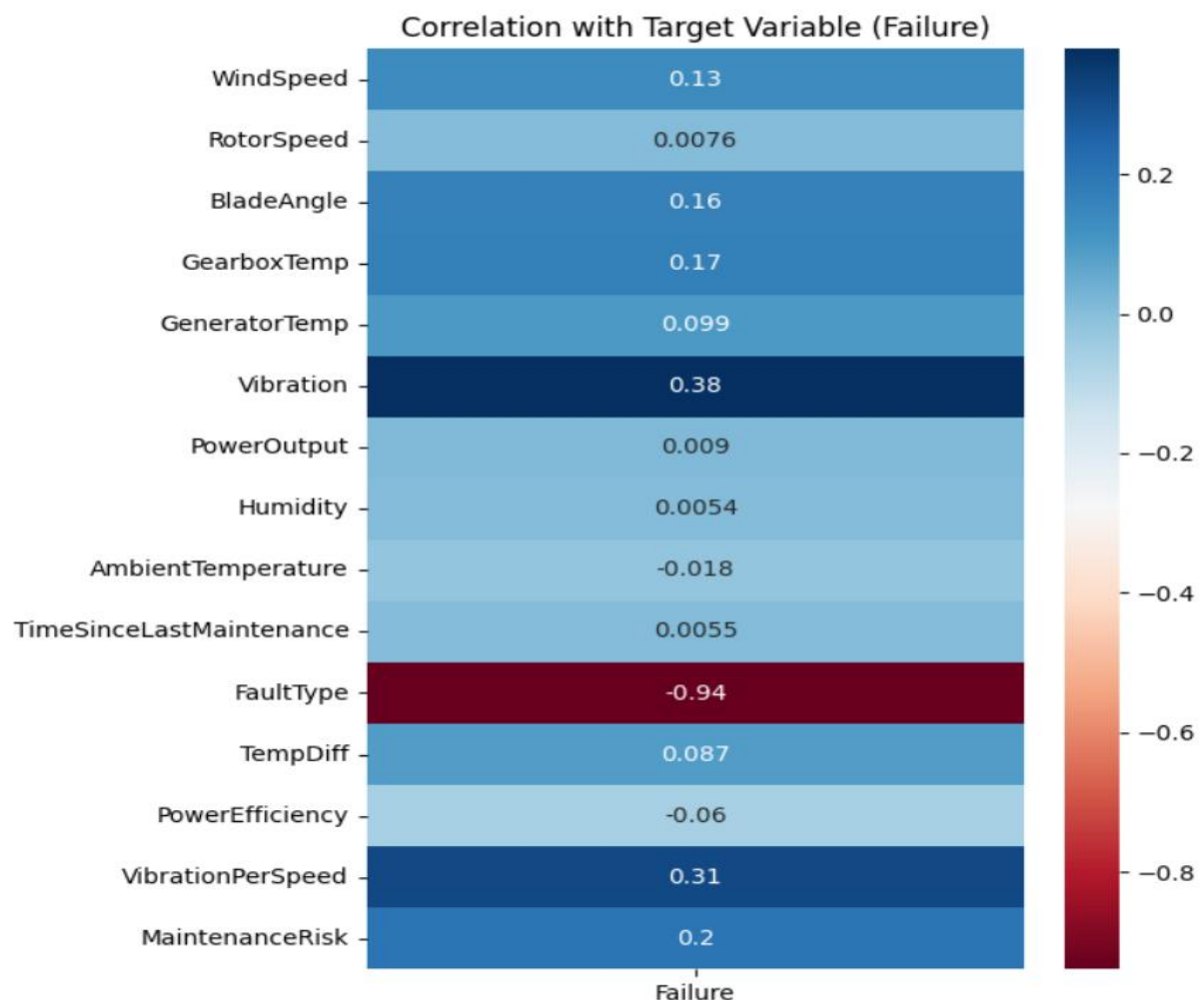
- The First step is Data exploration and cleaning. The goal of this phase is to ensure the dataset is free from errors, missing values, and irrelevant information.
- The dataset was loaded and examined to understand its structure, features, and data types.
- Missing values and duplicate records were checked and handled appropriately.
- Necessary preprocessing steps such as cleaning the data, encoding categorical values, and creating useful features were performed to make the data ready for model training.
- Created additional features such as Temperature Difference, Power Efficiency, Vibration risk and Maintenance Risk.
- **Tools Used:** Python Libraries Like Numpy, Pandas

2. Exploratory Data Analysis (EDA):

- EDA was carried out to understand the relationships between different turbine parameters and failure conditions.
- Statistical summaries and visualizations such as histograms, box plots, and correlation analysis were used to identify trends, patterns, and outliers in the dataset.
- Some of the Visualizations are:

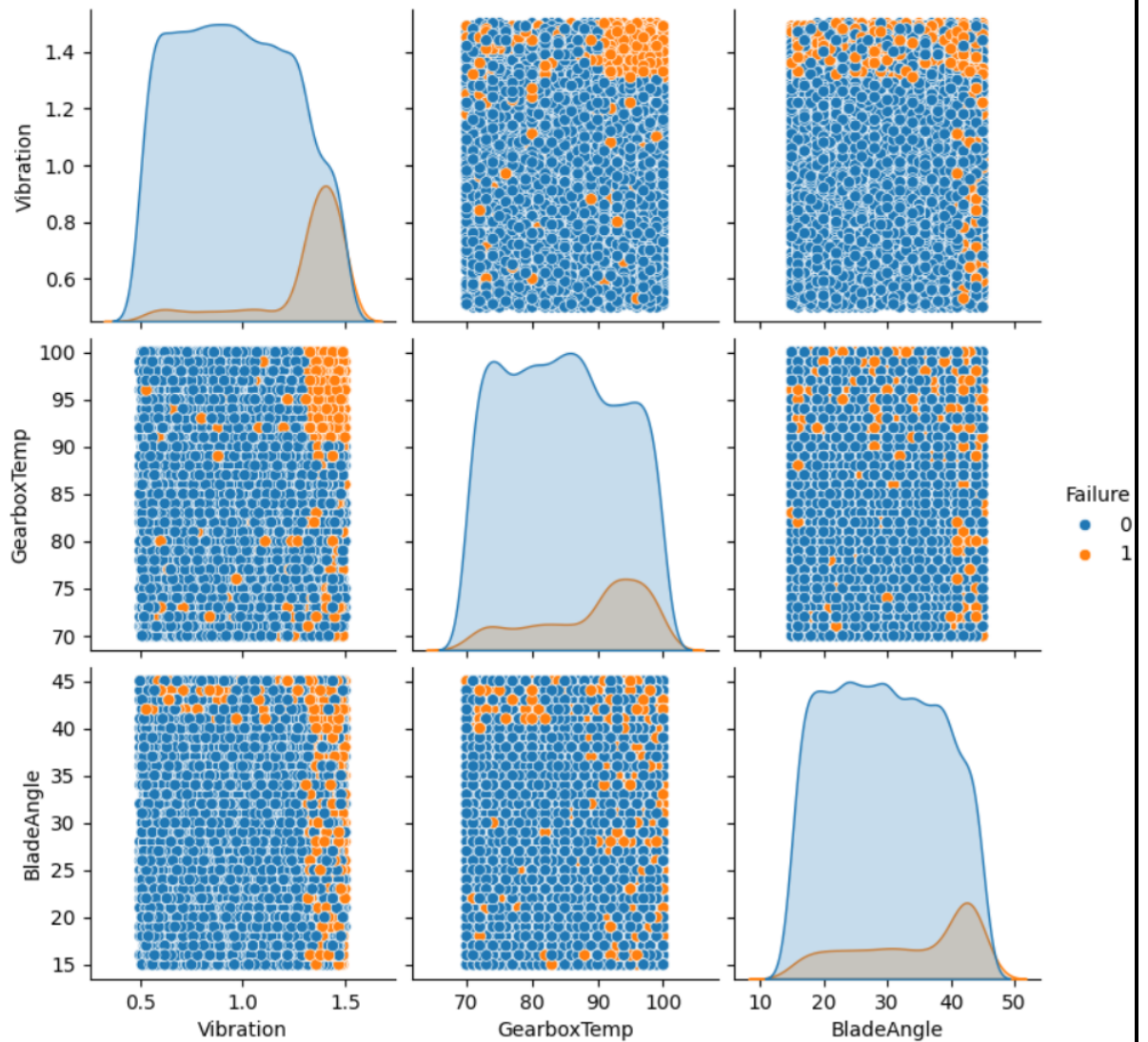
❖ Correlation Heatmap:

Used to identify relationships between features and detect which variables are strongly related to turbine failures.



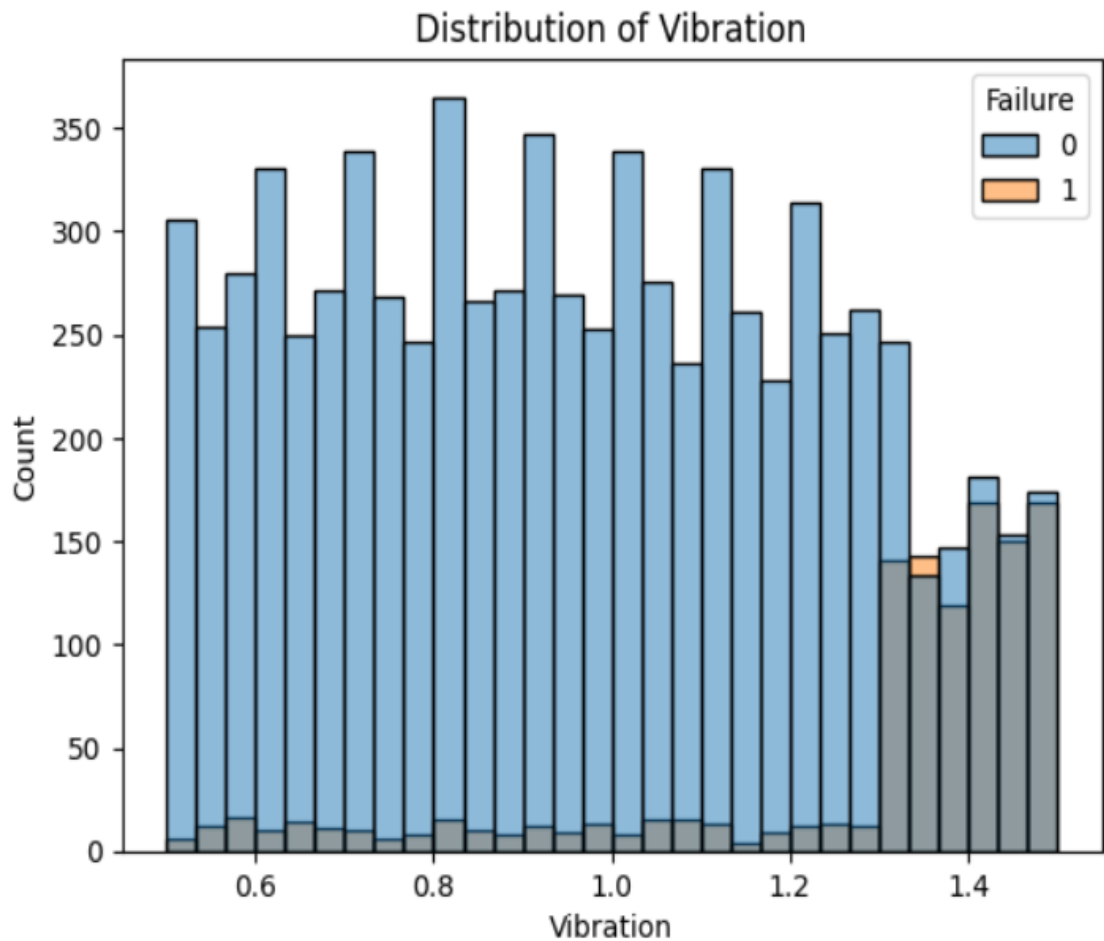
❖ Feature Interaction Analysis:

Analyzed pairwise relationships between vibration, gearbox temperature, and blade angle to identify patterns contributing to turbine failure.



❖ Vibration Distribution Analysis:

Visualized the distribution of vibration values to observe differences between failure and non-failure cases, showing that higher vibration levels are associated with turbine failure.



3. Model Selection and Training:

- In this stage, different machine learning models were selected and trained to predict turbine failures based on the processed sensor data.
- The dataset was divided into training and testing sets so that the model could learn from one portion of the data and be evaluated on unseen data.
- Multiple classification algorithms were tested to compare their performance and choose the best-performing model.
- Some of the Models are:

➤ **Logistic Regression:**

- Logistic Regression was used as a baseline classification model to predict whether a turbine would fail or not.
- It is simple, fast, and works well for binary classification problems.
- It helps in binary classification by estimating the probability of failure based on the input sensor features.

➤ **Decision Tree:**

- The Decision Tree model was used to classify turbine failures by creating a tree-like structure of decision rules.
- The model splits the data based on important features such as vibration, temperature, and speed values.

- This makes the prediction process easy to interpret and helps in identifying the conditions under which failures are more likely to occur.

➤ **Random Forest:**

- Random Forest was used as an ensemble learning technique that combines multiple decision trees to improve prediction performance.
- Each tree gives its own prediction, and the final output is based on majority voting.
- This model helps in reducing overfitting and provides more stable and accurate results compared to a single decision tree.

➤ **Gradient Boosting / XGBoost:**

- Gradient Boosting (or XGBoost, if used) was applied to further improve model accuracy.
- This model works by building multiple weak learners sequentially, where each new model corrects the errors made by the previous one.
- It is highly efficient for classification problems and is often used to achieve higher predictive accuracy.

4.Failure Type Prediction:

- After training and selecting the best-performing model, the next step was to predict the type of turbine failure.
- Instead of only identifying whether a failure may occur, the model was further used to classify the specific fault category.
- The prediction was performed based on important turbine parameters such as Wind Speed, Rotor Speed, Blade Angle, Gearbox Temperature, Generator Temperature, and Vibration.
- The model predicts different failure types such as:
 - ❖ **Gearbox Failure** – faults related to overheating, wear, or abnormal gearbox operation.
 - ❖ **Generator Failure** – issues caused by temperature rise or generator malfunction.
 - ❖ **Blade Failure** – faults related to blade angle, damage, or imbalance.
 - ❖ **Normal / No Failure** – indicates that the turbine is operating under healthy conditions.

5. Hyper Parameter Tuning:

- Hyperparameter tuning was performed to improve the performance and accuracy of the selected machine learning models.
- This process involves testing different parameter values and selecting the combination that gives the best results.

- Techniques such as GridSearchCV or RandomizedSearchCV were used to test multiple parameter combinations.
- This step helps in improving the model's generalization ability and achieving better prediction performance on unseen test data.
- The following parameters were tuned during this process:

- ❖ `n_estimators=50,100,150,200`

This defines the number of trees used in the Random Forest model. Increasing the number of trees generally improves accuracy and model stability.

- ❖ `max_depth=2,3,4,5`

Controls the maximum depth of each decision tree. Proper tuning helps in avoiding overfitting and improves generalization.

- ❖ `learning_rate=0.01,0.05,0.1,0.2`

Controls how much each tree contributes to the final model. Lower values make learning slower but more stable.

- ❖ `subsample=0.6,0.8,1.0`

Specifies the fraction of training samples used for each tree. Helps prevent overfitting by introducing randomness.

- ❖ `gamma=0,0.1,0.2`

Defines the minimum loss reduction required to make a split. Higher values make the model more conservative and reduce overfitting.

6. Post-Model Analysis:

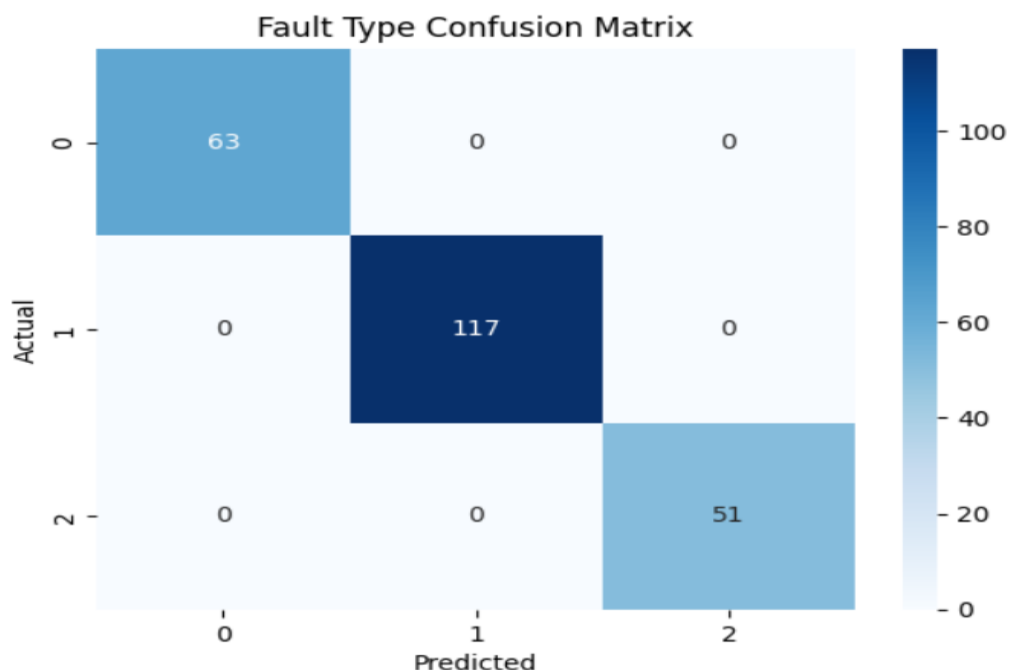
- After the model was trained and evaluated, post-model analysis was performed to understand its performance and reliability in predicting turbine failures.
- The Post Model Analysis was based upon some factors such as:

❖ Performance Evaluation:

- The trained models were analysed using metrics such as accuracy, precision, recall, and F1-score to measure their effectiveness in predicting failures.

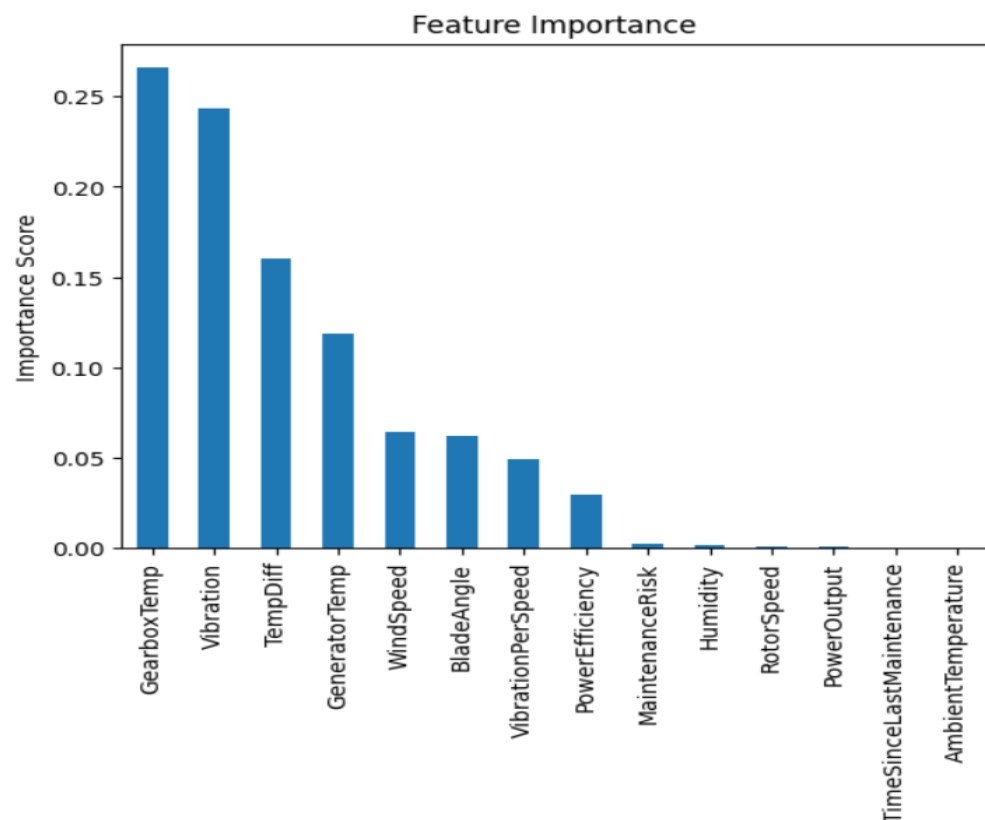
❖ Confusion Matrix Analysis:

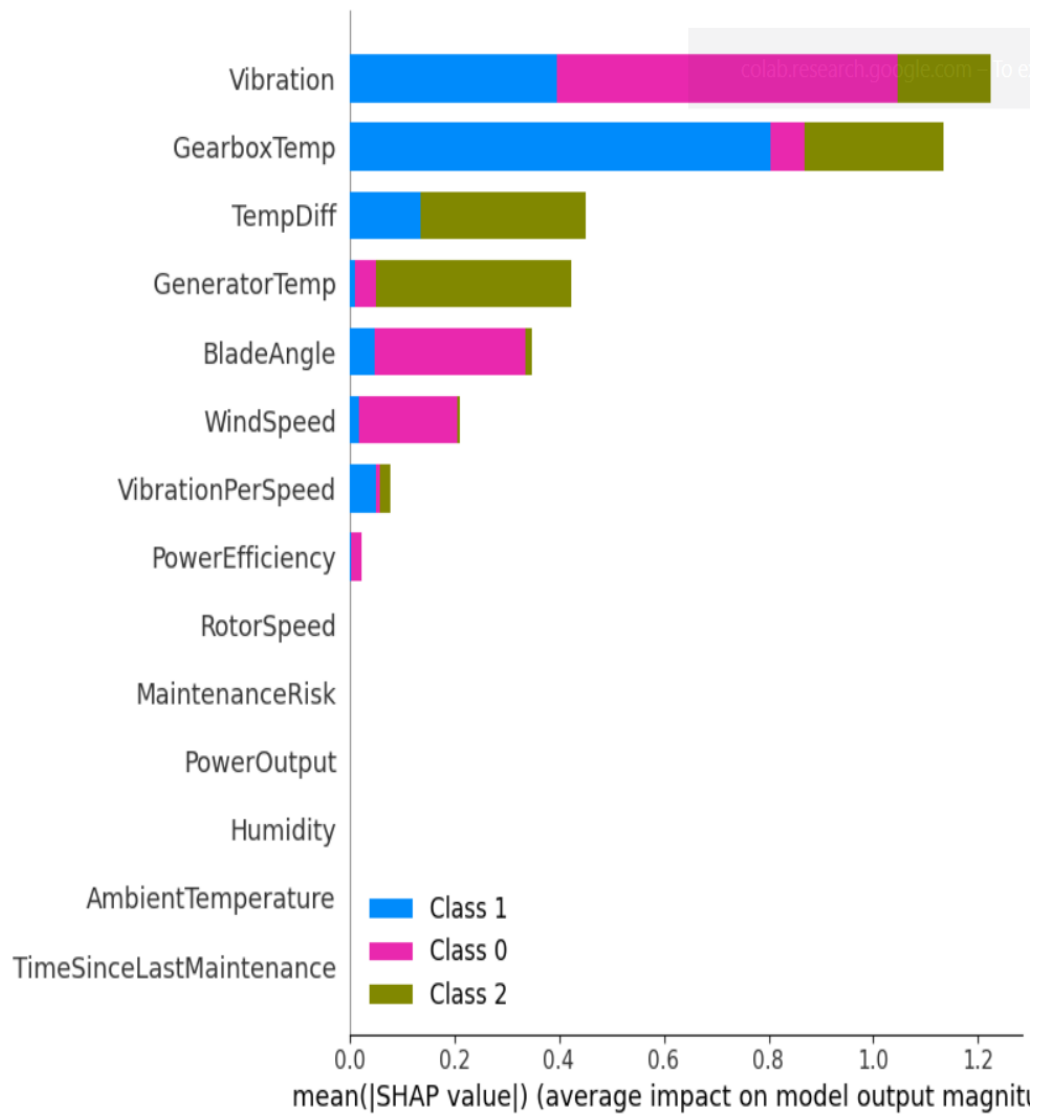
- A confusion matrix was used to compare the actual values with the predicted values.
- This helped in identifying the number of correct predictions, false positives, and false negatives.



❖ Feature Importance:

- The contribution of each feature such as Wind Speed, Rotor Speed, Gearbox Temperature, Generator Temperature, and Vibration was examined to determine which parameters had the highest impact on failure prediction.
- These features were examined by using certain tools like SHAP (SHapley Additive exPlanations).
- It helped in understanding how each feature contributed positively or negatively to the prediction output for individual samples as well as the overall model.





RESULTS

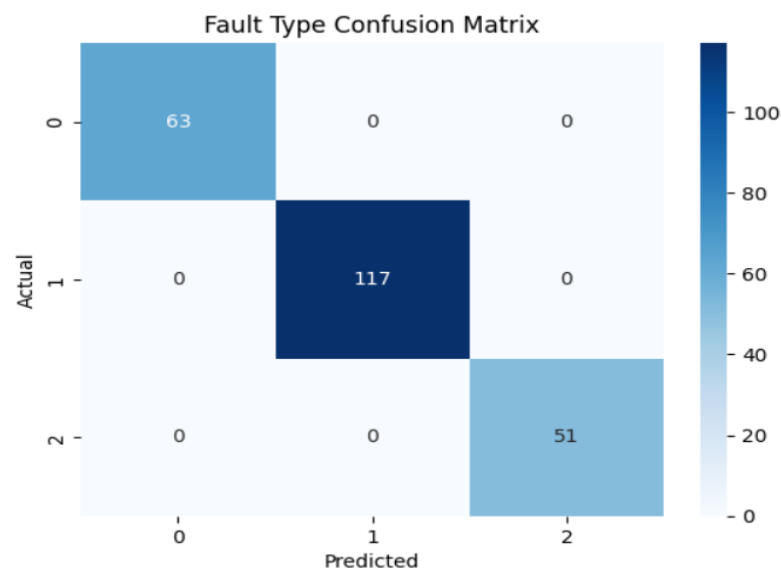
1. Model Accuracy:

- Logistic Regression achieved an accuracy of 79.3%, showing moderate performance but struggled with class imbalance.
- Decision Tree achieved an improved accuracy of 81.3%, providing better classification results compared to Logistic Regression.
- Random Forest achieved an accuracy of 90.1%, offering stable performance with reduced overfitting.
- XGBoost (XGB) achieved the highest accuracy 92.0% among all models, outperforming Logistic Regression, Decision Tree, and Random Forest, making it the best model for this project.

2. Confusion Matrix:

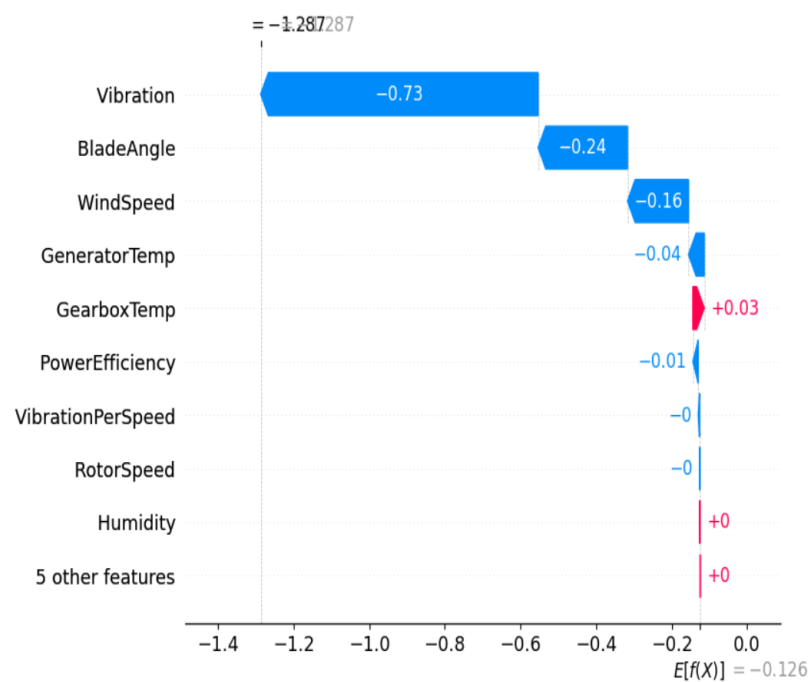
- For binary classification, the matrix shows how well the model distinguishes between failure (1) and non-failure (0) cases.
- For multi-class classification, the confusion matrix demonstrates accurate prediction of different fault types such as Class 0, Class 1, and Class 2.
- Most predictions fall along the diagonal, indicating that the model correctly classifies the majority of instances with

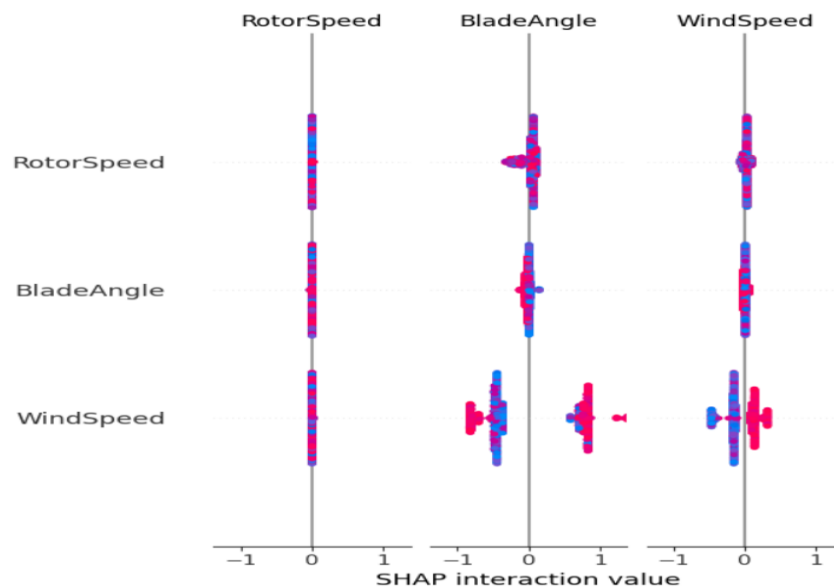
minimal misclassification.



3. SHAP Analysis:

- SHAP analysis revealed that Vibration, Blade Angle, Wind Speed, and Generator Temperature had significant influence on model predictions.
- Features like Vibration contributed strongly to predicting failure conditions.





4. Prediction Output:

- The trained model was tested with custom input values:

❖ InputParameters-1:

Wind Speed = 14.5, Rotor Speed = 15, Blade Angle = 29,
 Gearbox Temperature = 84, Generator Temperature = 84,
 Vibration = 0.9, Power Output = 1445, Humidity = 70,
 Ambient Temp = 25, Time Since Last Maintenance = 160

❖ PredictionResult-1:

The model successfully predicted “Turbine is Healthy”,
 demonstrating its ability to classify whether the turbine is
 Healthy/Fault.

```
Enter turbine values:
Wind Speed: 14.5
Rotor Speed: 15
Blade Angle: 29
Gearbox Temp: 84
Generator Temp: 84
Vibration: 0.9
Power Output: 1445
Humidity: 70
Ambient Temp: 25
Time Since Last Maintenance: 160

Prediction Result:
Turbine is Healthy
```

❖ Input Parameters-2:

Wind Speed = 19, Rotor Speed = 20, Blade Angle = 34,
Gearbox Temperature = 92, Generator Temperature = 95,
Vibration = 1.4, Power Output = 1540, Humidity = 70,
Ambient Temp = 30, Time Since Last Maintenance = 190

❖ Prediction Result-2:

The model successfully predicted “Failure Detected!”,
demonstrating its ability to classify whether the turbine is
Healthy/Fault.

```
Enter turbine values:  
Wind Speed: 19  
Rotor Speed: 20  
Blade Angle: 35  
Gearbox Temp: 92  
Generator Temp: 95  
Vibration: 1.4  
Power Output: 1540  
Humidity: 70  
Ambient Temp: 30  
Time Since Last Maintenance: 190  
  
Prediction Result:  
Failure Detected!
```

KEY FINDINGS

- The XGBoost (XGB) model achieved the highest accuracy(92.0%), outperforming Logistic Regression, Decision Tree, and Random Forest, making it the most suitable model for turbine failure prediction.
- Random Forest also showed strong performance with an accuracy of 90.1%, while Decision Tree (81.3%) and Logistic Regression (79.3%) showed comparatively lower accuracy.
- The models demonstrated high recall for failure cases, meaning most actual failures were successfully detected, which is critical for predictive maintenance.
- SHAP analysis revealed that Vibration, Blade Angle, Wind Speed, and Generator Temperature are the most influential features in predicting turbine failures.
- The model was able to accurately classify different fault types, such as generator, gearbox, and blade failures.
- The system successfully predicted real-time input data, correctly identifying a Generator Fault, proving its practical applicability.

CHALLENGES FACED

1. Data Imbalance Issue:

- The dataset had a large number of non-failure cases compared to failure cases, which made it difficult for models to correctly predict minority classes.
- The data imbalance issue was addressed by focusing on evaluation metrics like recall and F1-score instead of only accuracy.

2. Improving Model Performance:

- Multiple models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost were tested and compared.
- Finally, XGBoost was selected as it provided better accuracy and overall performance.

3. Feature Impact Understanding:

- Initially, it was difficult to understand which features were influencing the predictions.
- This challenge was solved using SHAP analysis, which helped identify important features like vibration and temperature.

CONCLUSION

This project successfully developed a machine learning-based system for turbine health monitoring and failure forecasting using key operational parameters. Multiple models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, were implemented and evaluated, with XGBoost achieving the highest accuracy and overall performance. The model effectively predicted both failure occurrence and specific fault types such as gearbox, generator, and blade failures. Important features like vibration and temperature were identified as major contributors, and SHAP analysis helped in understanding model predictions.

REFERENCES

- **Data Sources:** Wind Turbine Dataset
- **Tools Used:** Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, SHAP, GoogleColab)